

Question 1

- (a) Find $\lim_{x \rightarrow 0} \frac{\tan 3x}{2x}$ Marks
2
- (b) Find the acute angle (to nearest minute) between the lines : 2
 $2x - 3y - 1 = 0$ and $y = \frac{3x}{5} - 7$.
- (c) Divide the interval AB externally in the ratio 3:5 given the points $A(3, -2)$ and $B(-1, 7)$. 2
- (d) Expand and simplify $(2x + 3y)^4$ 2
- (e) Find the probability of getting 6 heads when a coin is tossed 8 times. 2
- (f) Write $7.\dot{1}\dot{2}$ as the sum of an infinite series. 2
 Hence write $7.\dot{1}\dot{2}$ as a mixed fraction.

Question 2.

The displacement x metres of a particle is given by :
 $x = 7 + 5 \sin 3t + 6 \cos 3t$ where t is the time in seconds.

- (a) Show that the particle moves in SHM, stating the centre of motion and period T . 4
- (b) Find the maximum speed of the particle. 2
- (c) Write $5 \sin 3t + 6 \cos 3t$ in the form $R \cos(3t - \alpha)$, where $R > 0$ and $0 < \alpha < 2\pi$. 2
- (d) Graph displacement x versus time t of the particle for $0 \leq t \leq 2\pi$. 2
- (e) Find the first time (to 2 decimal places) when the particle is 14 metres from the origin. 2

Question 3.

- (a) Evaluate $\int_0^4 \frac{2dx}{x^2 + 16}$ 2
- (b) (i) Factorise $x^3 + 2x^2 - 15x - 36$ 3
 (ii) Hence solve $x^3 + 2x^2 - 15x - 36 \geq 0$. 2
- (c) The velocity v of a particle is given by : 3
 $v = 5 + e^{-x}$ where x is the displacement of the particle.
 Find the displacement x as a function of time t if the particle is initially at the origin.
- (d) Find the rate of change $\frac{dF}{dt}$ (to 3 significant figures) if $F = G \frac{m_1 m_2}{r^2}$ 2
 where $G = 6.67 \times 10^{-11}$, $m_1 = 5.97 \times 10^{24}$, $m_2 = 1000$, $r = 1.5 \times 10^5$ and $\frac{dr}{dt} = 750$.

Question 4.

- (a) Find the area bounded by the lines $x = -1$, $x = -2$, the x -axis and the curve $y = \frac{1}{x}$. Marks
2
- (b) Find $\int \frac{4x + \sqrt{1-x^2}}{1-x^2} dx$ 2
- (c) Three engineers and nine councillors have a meeting around a circular table. If three councillors are between each engineer find number of possible seating arrangements. 2
- (d) Find the greatest coefficient of $(2x + 7)^{13}$. 3
- (e) The velocity v of a body is given by : $v = x \tan^2 x$, where x is the displacement. 3
 Find in simplest terms the acceleration $\ddot{x}$ of the body in terms of the displacement x .

Question 5.

- (a) Graph the curve $y = -2 \cos^{-1}\left(\frac{x}{3}\right)$. 3
- (b) Solve $\frac{4x-5}{2x+1} \leq 3$ 3
- (c) There are 8 red, 9 green and 6 yellow cards in a pack of cards. Five cards are drawn. Find the probability of obtaining 2 red and 3 green cards if it is known that at least one card is green. 2
 Leave the answer in $\frac{n}{r}$ form.
- (d) The point T lies on the inside of the acute angle XYZ . From T perpendiculars TV and TW are dropped to the angle arms YX and YZ respectively. From point Y , the perpendicular YN is dropped to the interval VW .
 (i) Draw a diagram showing all the information. 1
 (ii) Prove that $\angle VYN = \angle TYW$. 3

Question 6.

- (a) Using the substitution $x = \frac{1}{y}$ and integration tables find $\int \frac{dx}{x\sqrt{1-x^2}}$. 4
- (b) Prove by Mathematical Induction : 4
 $1 \times 3 \times 5 \times \dots \times (2n-1) = \frac{(2n)!}{2^n n!}$
- (c) A man takes out a loan for \$260 000 to be paid in equal monthly payments over 25 years. If the interest on the loan is 8 %p.a. monthly reducible, find the monthly repayment R . 4

Question 7.

- | | | Marks |
|-----|--|--------------|
| (a) | (i) Show that $T = A + Be^{-kt}$ is a solution of $\frac{dT}{dt} = -k(T - A)$. | 2 |
| | (ii) A barbeque plate is heated to $85^{\circ}C$ when the ambient temperature is $22^{\circ}C$.
The plate cools to $70^{\circ}C$ in 16 minutes.
Assuming Newton's Law of Cooling find the time for the plate to cool to $30^{\circ}C$. | 4 |
| (b) | A projectile is fired with initial speed V m/s from the origin O at an angle of α to the horizontal ($0 \leq \alpha < 90^{\circ}$).

The trajectory equation is given by : $y = x \tan \alpha - \frac{gx^2}{2V^2}(1 + \tan^2 \alpha)$.

The projectile reaches a maximum height, and on the downward motion the projectile hits the target 20 metres above ground level at an angle of 27° to the horizontal.
Find the horizontal distance R that the target is from the Origin O (to nearest cm),
if the angle of projection α is 45° and the acceleration due to gravity g is $10m/s^2$. | 4 |
| (c) | The sequences $\{1, 3, 5, \dots, p\}$ and $\{1, 3, 5, \dots, q\}$ contain the integer values of p and q respectively.

Find the value of $p+q$ if :

$(1+3+5+\dots+p) + (1+3+5+\dots+q) = (1+3+5+\dots+33)$ | 2 |

End of Exam

$$(a) \lim_{x \rightarrow 0} \frac{\tan 3x}{2x} = \lim_{x \rightarrow 0} \frac{\sec 3x}{2} \cdot \frac{3}{2 \cos 3x}$$

$$= 1 \times \frac{3}{2 \times 1}$$

$$= \frac{3}{2}$$

$$m_1 = \frac{2}{3} \quad m_2 = \frac{3}{5}$$

$$\therefore \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{2}{3} - \frac{3}{5}}{1 + \frac{2}{3} \cdot \frac{3}{5}} \right|$$

$$= \left| \frac{10 - 9}{15 + 6} \right|$$

$$= \left| \frac{1}{21} \right|$$

$$\theta = 2^\circ 44'$$

$$3:5$$

$$A(3, -2) \times B(-1, 7)$$

$$P = \left(\frac{-3 - 15}{3 - 5}, \frac{21 + 10}{3 - 5} \right)$$

$$= (9, -15\frac{1}{2})$$

$$(d) (2x + 3y)^4 = 16x^4 + 96x^3y + 216x^2y^2 + 216xy^3 + 81y^4$$

$$(e) P(6H) = {}^8C_6 \cdot \left(\frac{1}{2}\right)^8$$

$$= \frac{28}{256}$$

$$= \frac{7}{64}$$

$$(f) 7.12 = 7 + 0.12 \times 100 = 7 + \frac{12}{100}$$

$$= 7 + \frac{0.12}{1 - \frac{1}{100}}$$

$$= 7\frac{4}{33}$$

Ext 1.

2009.

Use Soln.

$$2(a) x = 7 + 5 \sin 3x + 6 \cos 3x$$

$$v = 15 \cos 3x - 18 \sin 3x$$

$$\ddot{x} = -45 \sin 3x - 54 \cos 3x$$

$$= -9 [5 \sin 3x + 6 \cos 3x]$$

$$= -9 [7 + 5 \sin 3x + 6 \cos 3x - 7]$$

$$\ddot{x} = -9 [x - 7]$$

which is of the form

$$\ddot{x} = -n^2 (x - B)$$

where $n^2 = 9$

$$n = 3$$

(b) Centre motion $x = 7m$.

$$\text{Period } T = \frac{2\pi}{3} \text{ s}$$

$$(c) v_{\max} = \sqrt{15^2 + 18^2}$$

$$= 3\sqrt{61} \text{ m/s}$$

$$(c) 5 \sin 3x + 6 \cos 3x = R \cos(3x - \alpha)$$

$$= R \cos 3x \cos \alpha + R \sin 3x \sin \alpha$$

$$\therefore R \cos \alpha = 5 \quad R \sin \alpha = 6 \quad 0 < \alpha < \frac{\pi}{2}$$

$$R \cos \alpha = 5$$

$$R \sin \alpha = 6$$

$$R = \sqrt{5^2 + 6^2}$$

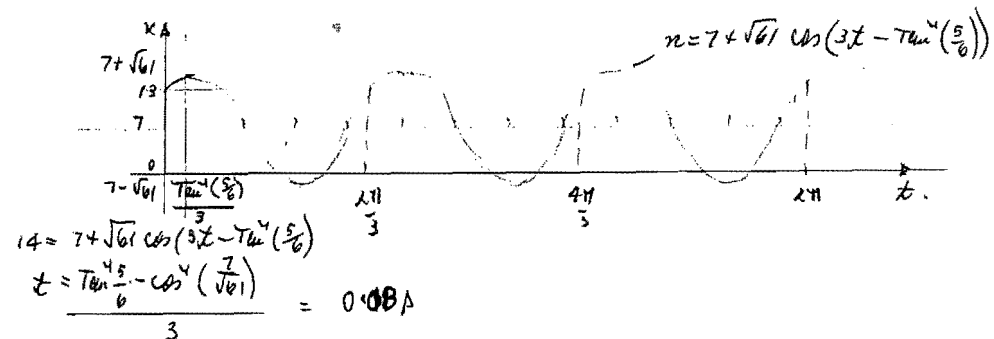
$$= \sqrt{61}$$

$$\tan \alpha = \frac{6}{5}$$

$$\alpha = \tan^{-1}\left(\frac{6}{5}\right)$$

$$\therefore 5 \sin 3x + 6 \cos 3x = \sqrt{61} \cos\left(3x - \tan^{-1}\left(\frac{6}{5}\right)\right)$$

(d)



$$\begin{aligned} \int_0^{\pi/4} \frac{2 dx}{x^2 + 16} &= \frac{2.1}{4} \left[\tan^{-1} \frac{x}{4} \right]_0^{\pi/4} \\ &= \frac{1}{2} \left(\tan^{-1} 1 - 0 \right) \\ &= \frac{\pi}{8} \end{aligned}$$

6) Let $P(x) = x^3 + 2x^2 - 15x - 36$

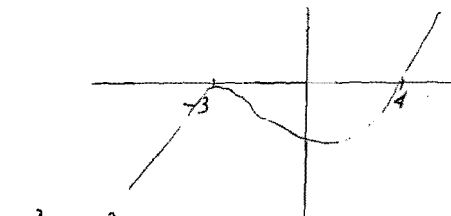
$$\begin{aligned} P(1) &= -48 \\ P(-1) &= -20 \\ P(2) &= -30 \\ P(-2) &= -6 \\ P(3) &= -36 \\ P(-3) &= -27 + 18 + 45 - 36 \\ &= 0 \end{aligned}$$

$\therefore x+3$ is a factor of $P(x)$

$$\begin{aligned} \therefore P(x) &= (x+3)(x^2 - x - 12) \\ &= (x+3)(x+3)(x-4) \end{aligned}$$

$$\therefore x^3 + 2x^2 - 15x - 36 = (x+3)^2(x-4)$$

(ii)



$$x^3 + 2x^2 - 15x - 36 > 0$$

$$x < -3 \text{ or } x > 4.$$

3(c)

$$v = 5 + e^{-u}$$

$$\frac{dv}{du} = 5 + e^{-u}$$

$$\frac{dt}{dv} = \frac{1}{5 + e^{-u}}$$

$$= \frac{e^u}{5e^u + 1}$$

$$\therefore t = \frac{1}{5} \ln(5e^u + 1) + C$$

$$\text{But } t=0 \text{ when } u=0 \Rightarrow C = -\frac{1}{5} \ln 6$$

$$\therefore t = \frac{1}{5} \ln \left(\frac{5e^u + 1}{6} \right)$$

$$u = \ln \left(\frac{6e^{5t} - 1}{5} \right)$$

(d)

$$F = G \frac{m_1 m_2}{r^2}$$

$$\frac{dF}{dt} = -2G \frac{m_1 m_2}{r^3} \frac{dr}{dt}$$

$$= \frac{-2 \times 6.67 \times 10^{-11} \cdot 5.97 \times 10^{24} \times 10^3 \times 750}{(1.5 \times 10^5)^3}$$

$$= -1.77 \times 10^5$$

$$\frac{4}{(8)} \quad \text{Area} = \int_{-1}^{-2} \frac{1}{x} dx$$

$$= \left[\ln(-x) \right]_{-1}^{-2}$$

$$= \underline{\ln 2} \cdot \text{unit}^2$$

$$\int \frac{4x + \sqrt{1-x^2}}{1-x^2} dx = \int \frac{4x}{1-x^2} + \frac{1}{\sqrt{1-x^2}} dx$$

$$= -2 \ln(1-x^2) + \sin^{-1} x + C$$

Ways E = 2!

Ways C = 9!

Total ways = 2! × 9!

$$T_{r+1} = {}^{13}C_r (2x)^{13-r} 7^r$$

$$T_r = {}^{13}C_{r-1} (2x)^{14-r} 7^{r-1}$$

$$\frac{T_{r+1}}{T_r} = \frac{13!}{r! (13-r)!} \cdot \frac{(r-1)! (14-r)!}{13!} \cdot \frac{2}{2^{14-r}} \cdot \frac{7}{7^{r-1}} \cdot \frac{x}{x^{14-r}}$$

$$= \frac{14-r}{r} \cdot \frac{7}{2} \cdot \frac{1}{x}$$

For greatest coefficient $\frac{T_{r+1}}{T_r} \geq 1$

$$\frac{7(14-r)}{2r} \geq 1$$

$$98 - 7r \geq 2r$$

$$9r \leq 98$$

$$r \leq 10 \frac{8}{9}$$

$\Rightarrow r \leq 10$

$T_{11} > T_{10}$
 $T_{10} > T_9$
 $T_9 > T_8$

$T_{12} < T_{11}$
 $T_{13} < T_{12}$
 $T_{14} < T_{13}$

greatest coefficient $r=10$

$$T_{11} = {}^{13}C_{10} 2^{13-10} 7^{10}$$

$$= 2288 \times 7^{10}$$

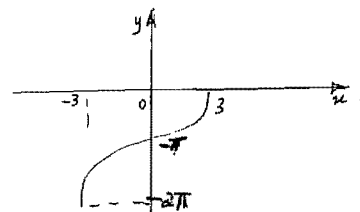
$$(e) \quad \dot{x} = v \frac{dv}{dx}$$

$$= x \tan^2 x \frac{d}{dx} [x \tan^2 x]$$

$$= x \tan^2 x \left[\tan^2 x + 2 \tan x \sec^2 x \right]$$

$$= x \tan^3 x \left[\tan x + 2 \sec^2 x \right]$$

5/ (a)



$$(d) \quad \frac{4x-5}{2x+1} \leq 3 \quad x \neq -\frac{1}{2}$$

$$(2x+1)(4x-5) \leq 3(2x+1)^2$$

$$(2x+1)(4x-5-6x-3) \leq 0$$

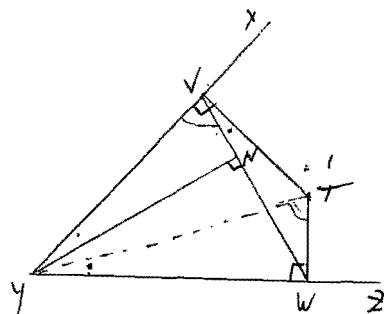
$$(2x+1)(-2x-8) \leq 0$$

$$(2x+1)(x+4) \geq 0$$

$$x \leq -4 \quad \text{OR} \quad x > -\frac{1}{2}$$

$$(c) \quad P(2A, 3G) = \frac{{}^8C_2 \cdot {}^9C_3}{{}^{23}C_5 - {}^{14}C_5}$$

3(d)



(i) ✓

$$(ii) \angle YVT + \angle YWT = 90^\circ + 90^\circ = 180^\circ$$

$\therefore$ $\angle YVTW$ is a cyclic quadrilateral
(opposite angles are supplementary) -

In ΔYTW and ΔYVN

$$\angle YTW = \angle YVN \text{ (Angles at the circumference in the same segment are equal)} -$$

$$\angle YWT = \angle YNV \text{ (Both right angles)} -$$

$\therefore \Delta YTW \sim \Delta YVN$ (Equiangular)

$\therefore \angle VYN = \angle YW$ (Corresponding angles of similar triangles are equal)

(e)

$$x = \frac{1}{y}$$

$$dx = -\frac{1}{y^2} dy$$

$$i \int \frac{dx}{x \sqrt{1-x^2}} = \int \frac{-1}{y^2} \cdot \frac{dy}{\frac{1}{y} \sqrt{1-\frac{1}{y^2}}}$$

$$= \int \frac{-1}{y^2} \cdot \frac{y}{\sqrt{y^2-1}} dy$$

$$= - \int \frac{dy}{\sqrt{y^2-1}}$$

$$= - \ln(y + \sqrt{y^2-1})$$

$$= - \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2}-1}\right)$$

$$= \ln\left(\frac{x}{1 + \sqrt{1-x^2}}\right) + C$$

5(b) Step 1.

$$n=1 \quad LHS=1 \quad RHS = \frac{(2n)!}{2^n n!}$$

$$= \frac{2!}{2 \cdot 1!}$$

$$= \frac{2}{2}$$

$$= 1$$

$\therefore LHS = RHS$ True $n=1$

Step 2 Assume statement is true $n=k$

$$1 \times 3 \times 5 \times \dots \times (2k-1) = \frac{(2k)!}{2^k k!}$$

To prove statement is true $n=k+1$

$$1 \times 3 \times 5 \times \dots \times (2k-1)(2k+1) = \frac{[2(k+1)]!}{2^{k+1} (k+1)!}$$

$$\text{Now } 1 \times 3 \times \dots \times (2k-1)(2k+1) = \frac{(2k)!}{2^k k!} \cdot (2k+1) \quad (\text{By assumption})$$

$$= \frac{(2k)! (2k+1)(2k+2)}{2^k k! 2k+2}$$

$$= \frac{(2k+2)!}{2^k k! 2 \cdot (k+1)}$$

$$= \frac{[2(k+1)]!}{2^{k+1} (k+1)!}$$

$\therefore$ If statement true $n=k$ it is also true $n=k+1$.
Since statement is true $n=1$ it is also true $n=1+1=2$,
 $n=2+1=3$ and so on for all positive integers n .

6(c) monthly interest $= \frac{0}{1200} = \frac{1}{150}$

$n = 25 \times 12$
 $= 300$

$R \equiv$ Repayment

Amount owing end 1st month $= 260000 \times (1 + \frac{1}{150}) - R$

Amount owing end 2nd month $= [260000 (1 + \frac{1}{150}) - R] (1 + \frac{1}{150}) - R$

$$= 260000 \cdot (\frac{151}{150})^2 - R [1 + \frac{151}{150}]$$

Amount owing end 3rd month $= [260000 (\frac{151}{150})^2 - R [1 + \frac{151}{150}]] (\frac{151}{150}) - R$

$$= 260000 (\frac{151}{150})^3 - R [1 + \frac{151}{150} + (\frac{151}{150})^2]$$

end 300 months $0 = 260000 (\frac{151}{150})^{300} - R [1 + (\frac{151}{150}) + (\frac{151}{150})^2 + \dots + (\frac{151}{150})^{299}]$

$$R \left[\frac{(\frac{151}{150})^{300} - 1}{\frac{151}{150} - 1} \right] = 260000 (\frac{151}{150})^{300}$$

$$R = \frac{260000 \cdot \frac{1}{150} \cdot (\frac{151}{150})^{300}}{(\frac{151}{150})^{300} - 1}$$

Repayment $= \$2006.72$ per month.

3) (i) $T = A + Be^{kt}$
 $\frac{dT}{dt} = -kBe^{kt}$
 $= -k[T - A]$
 but $Be^{kt} = T - A$ — (2)

ii) $t=0$ $T=85$ $A=22$
 $t=16$ $T=70$

$\therefore T = 22 + Be^{kt}$ — (1)

$85 = 22 + B$

$B = 63$

$T = 22 + 63e^{kt}$

$70 = 22 + 63e^{-16k}$

$k = \frac{1}{16} \ln\left(\frac{48}{63}\right)$

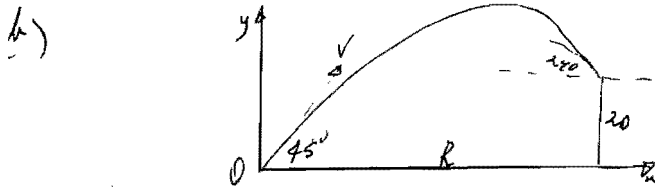
$\therefore T = 22 + 63e^{-\frac{t}{16} \ln\left(\frac{63}{48}\right)}$

$30 = 22 + 63e^{-\frac{t}{16} \ln\left(\frac{63}{48}\right)}$

$e^{-\frac{t}{16} \ln\left(\frac{63}{48}\right)} = \frac{8}{63}$

$t = \frac{-\ln\left(\frac{8}{63}\right)}{\ln\left(\frac{63}{48}\right)}$

$= 121 \text{ mins.}$



$y = x \tan \theta - \frac{gx^2}{2v^2} (1 + \tan^2 \theta)$

$= x - \frac{5x^2}{v^2} (1 + 1)$

$y = x - \frac{10x^2}{v^2}$

$20 = R - \frac{10R^2}{v^2}$ — (1)

Also $\frac{dy}{dx} = \tan \theta - \frac{gx}{v^2} (1 + \tan^2 \theta)$

$-\tan 27^\circ = 1 - \frac{10R}{v^2} \cdot 2$

$\frac{R}{v^2} = \frac{1 + \tan 27^\circ}{20}$ — (2)

$\therefore$ From (1) $20 = R - 10R \cdot \left[\frac{1 + \tan 27^\circ}{20} \right]$ — (1)

$R = \frac{20}{1 - \frac{1}{2} - \frac{1}{2} \tan 27^\circ}$
 $= \frac{40}{1 - \tan 27^\circ}$

Range $R = 81.55 \text{ m.}$ — (1)

(c) $(1+3+5+\dots+p) \times (1+3+5+\dots+q) = 1+3+5+\dots+33$

New $2n-1 = p$

Number of terms $n = \frac{p+1}{2}$

$\therefore \frac{1}{2} \left(\frac{p+1}{2} \right) (1+p) \times \frac{1}{2} \left(\frac{q+1}{2} \right) (1+q) = \frac{1}{2} \cdot \frac{33+1}{2} \cdot (33+1)$
 $\left(\frac{p+1}{2} \right)^2 + \left(\frac{q+1}{2} \right)^2$

$\therefore \frac{p+1}{2} = 15$ $\frac{q+1}{2} = 17$

$p = 29$

$q = 33$

$\therefore \underline{p+q = 62}$